

Indian Mutual Fund Quantitative Analytics & Risk Intelligence Platform

A Production-Grade Data Engineering Warehouse, Multi-Factor Performance Scorecard, Advanced Tail-Risk Engine (VaR/CVaR, Monte Carlo, Markowitz Frontier), and Executive Business Intelligence System

Author / Candidate:	Varundeep (Full-Stack Data Engineering & Quantitative Analytics)
Institutional Partner:	Bluestock Fintech — Capstone Project Division
Project Repository:	github.com/Varundeep09/bluestock-mf-capstone
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Primary Tech Stack:	Python 3.13, SQLite 3, SQLAlchemy, Pandas, SciPy, Power BI Desktop, ReportLab

Executive Highlights & Core Results:

- Automated End-to-End Pipeline:** Automated 3-stage ETL orchestrating 10 raw CSVs and 6 live AMFI API feeds into an 11-table star-schema database in 25.4s.
- Market Performance Scorecard:** 40-fund multi-factor ranking identifying **Kotak Flexicap Fund (Score 71.38)** and **SBI Small Cap Fund (Score 70.25)** as industry leaders.
- Structural SIP Liquidity Surge:** Domestic monthly SIP inflows expanded by **+169.2%** (■11,517 Cr to ■31,002 Cr), buffering Indian equities against global volatility.
- Advanced Quantitative Risk Models:** 95% Historical VaR/CVaR, rolling 90-day Sharpe dynamics, 5-year Monte Carlo GBM projections, and Markowitz portfolio optimization.

1. Executive Summary

The Indian mutual fund ecosystem has undergone structural transformation over the 2022–2026 observation window, shifting from an institutionally dominated marketplace into a mass-retailized wealth creation engine. This Capstone Project delivers an enterprise-grade quantitative analytics platform and data warehouse engineered to ingest, clean, validate, and analyze complex mutual fund performance datasets, investor transactions, and macroeconomic flows.

Core Engineering & Analytical Deliverables:

- **Automated Master ETL Pipeline (scripts/etl_pipeline.py):** Integrates 10 raw operational datasets and 6 live AMFI API feeds, executing schema validation, date parsing, duplicate removal, forward-fill NAV alignment, and loading into an 11-table SQLite database in 25.4 seconds.
- **Multi-Factor Performance Scorecard (0–100 Scale):** Rigorously weights trailing 3Y CAGR (30%), Sharpe ratio (25%), Jensen's Alpha (20%), inverted Total Expense Ratio (15%), and inverted Maximum Drawdown (10%) across 40 schemes.
- **Advanced Risk & Customer Analytics:** Implements 95% Historical Value at Risk (VaR), Conditional VaR (Expected Shortfall), rolling 90-day Sharpe ratios, investor cohort retention, and Herfindahl-Hirschman Index (HHI) sector concentration flags.
- **Modern Portfolio Theory & Stochastic Cones:** Models Markowitz Efficient Frontier optimal tangency weights and 1,000-path Geometric Brownian Motion (GBM) Monte Carlo NAV simulations over a 5-year forward horizon.
- **Executive Business Intelligence Dashboard:** 4-page interactive Power BI interface deployed with complete KPI cards, demographic breakdowns, and portfolio exposure heatmaps.

Headline Empirical Business Findings:

Metric Dimension	Empirical Finding / Value	Strategic Business Implication
Monthly SIP Growth	■11,517 Cr -> ■31,002 Cr (+169.2%)	Provides domestic liquidity cushion against foreign institutional outflows.
Equity Folio Share	69.98% of 26.12 Cr Total Folios	Rapid financialization of household savings into equity capital markets.
Top Ranked Fund	Kotak Flexicap (Score: 71.38/100)	Optimal multi-factor balance of alpha (1.85), Sharpe (0.98), and downside defense.
Direct Plan Cost Alpha	60 to 110 bps Expense Savings	Direct plans generate ~0.8% higher annualized net CAGR for long-term investors.
Empirical SIP Gap	Mean: 64.9 Days Median: 64.7 Days	97.8% flagged under rigid 35-day rule; calls for threshold recalibration to 65 days.
Sector Concentration	4 Funds Flagged with HHI > 2500	Axis Bluechip leads HHI (2967.7) due to heavy 48.7% IT equity allocation.

2. Data Sources & Architecture Catalog (Part 1)

The analytical data warehouse unifies 10 core mutual fund operational datasets covering temporal granularities from daily NAV sessions to quarterly AUM disclosures, complemented by real-time automated API pulls from AMFI India.

Comprehensive Dataset Inventory:

Dataset File	Target DB Table	Rows	Key Attributes & Contents
01_fund_master.csv	dim_fund	40	AMFI code, scheme name, AMC, category, plan, launch date, TER, risk grade.
02_nav_history.csv	fact_nav	46,000	Daily NAV history across 40 schemes over 1,150 trading sessions (2022–2026).
03_aum_by_fund_house.csv	fact_aum	90	Quarterly AUM disclosures in ₹ Lakh Cr and ₹ Cr across leading AMCs.
04_monthly_sip_inflows.csv	fact_sip_industry	48	Industry monthly SIP inflows (₹ Cr), active accounts, and YoY growth rates.
05_category_inflows.csv	fact_category_inflows	144	Monthly net capital inflows across 12 SEBI categories (Large, Mid, Small, Debt).
06_industry_folio_count.csv	fact_folio_count	21	Quarterly folio expansion segmented by Equity, Debt, Hybrid, and Index folios.
07_scheme_performance.csv	fact_performance	40	Trailing 1Y/3Y/5Y returns, Alpha, Beta, Sharpe, Sortino, Max Drawdown, Ratings.
08_investor_transactions.csv	fact_transactions	32,778	Granular transaction logs across SIP, Lumpsum, Redemptions, Demographics, KYC.
09_portfolio_holdings.csv	fact_portfolio	322	Stock ticker, company name, sector classification, allocation weights, market value.
10_benchmark_indices.csv	fact_benchmark	8,050	Daily closing values for NIFTY50, NIFTY100, NIFTY500, NIFTY_MIDCAP150, etc.
api_*.csv (6 Schemes)	Raw Staging	20,208	Real-time live NAV pulls from mfapi.in for HDFC Top 100, SBI, ICICI, etc.

Live AMFI API Fetch Integration (scripts/live_nav_fetch.py):

To augment static raw CSVs with dynamic market feeds, an automated REST ingestion pipeline interfaces with the https://api.mfapi.in/mf/{amfi_code} endpoint. It fetches live historical NAV series for 6 core benchmark funds (HDFC Top 100, SBI Small Cap, ICICI Liquid, Nippon India Small Cap, Kotak Emerging Equity, and Mirae Asset Large Cap). The script parses nested JSON payloads, converts dates to ISO-8601 strings, and caches clean staging files into `data/raw/api_{amfi_code}.csv` with zero manual intervention.

2. Data Sources & Architecture Catalog (Part 2)

Data Authenticity & Verification Boundary Disclosures:

To maintain complete academic transparency and fiduciary integrity, all data elements ingested into the Bluestock Analytics Platform are categorized according to their authentic source provenance:

Data Domain	Data Provenance & Nature	Verification & Ground Truth Status
Fund Metadata	Real AMFI Disclosures	Scheme names, AMFI codes, AMC brands, expense ratios (TER), and fund manager assignments reflect authentic regulatory filings.
NAV Time Series	Anchored Market Series	Daily NAVs spanning 2022–2026 across 40 schemes are anchored to actual mutual fund return trajectories and daily trading volatility.
Benchmark Indices	NSE Historical Series	Daily closing indices for NIFTY 50, NIFTY 100, NIFTY 500, and NIFTY MIDCAP 150 reflect real Indian equity market movements.
AUM & SIP Inflows	AMFI Industry Aggregates	Monthly macroeconomic SIP inflows (₹11,517 Cr -> ₹31,002 Cr) and quarterly AUMs mirror official AMFI press releases.
Investor Demographics	Synthetic Transaction Layer	32,778 transaction records were synthetically modeled to match real Indian demographic distributions (T30/B30, age, payment rails).

Key Data Cleaning & Normalization Decisions:

- Temporal Standardization:** All date strings across heterogeneous raw CSV formats (DD-MM-YYYY, YYYY-MM-DD, DD/MM/YYYY) were parsed to unified ISO-8601 YYYY-MM-DD timestamps.
- Trading Day Forward-Filling:** Weekend and public holiday gaps in daily NAV series were strictly forward-filled within active scheme timelines, while preventing synthetic leakage across pre-inception fund gaps.
- Boundary Validation & Outlier Clamping:** Strict range checks were enforced on financial metrics: Total Expense Ratios (TER) bounded between 0.10% and 2.50%; daily returns bounded within ±20%; negative NAV values rejected.
- Transaction Standardization:** Transaction types were normalized into standardized enumerations (SIP, LUMPSUM, REDEMPTION, SWITCH) with ticket sizes bounded between ₹100 and ₹5,000,000.

3. System Architecture & Data Engineering (Part 1)

The Bluestock Analytics Engine is built upon a decoupled 5-layer pipeline architecture designed for automated execution, idempotent staging, strict relational schema constraints, and sub-second quantitative query performance.

5-Layer Pipeline Architecture:

Architecture Layer	Components & Technology	Function & Responsibilities
1. Ingestion Layer	scripts/data_ingestion.py scripts/live_nav_fetch.py	Validates raw file presence, inspects column schemas, and fetches live AMFI API data without altering raw files.
2. Processing & Cleaning	scripts/data_cleaning.py Pandas, NumPy	Parses dates, handles nulls, validates numeric bounds (TER 0.1%–2.5%), standardizes transaction types, computes daily returns.
3. Warehouse Layer	data/db/bluestock_mf.db SQLite 3, SQLAlchemy	Star schema with 11 tables, strict primary keys, foreign key constraints (PRAGMA foreign_keys = ON), and B-tree indexes.
4. Quantitative Engine	scripts/compute_metrics.py scripts/advanced_analytics.py	Computes CAGR, Sharpe, Sortino, Alpha, Beta, VaR/CVaR, cohorts, churn, HHI, Monte Carlo paths, and Markowitz frontier.
5. Business Intelligence	Power BI Desktop dashboard/bluestock_mf.pbix	Interactive 4-page dashboard featuring slicers, dynamic KPI cards, demographic funnels, and risk-adjusted return scatter matrices.

Master Orchestration Workflow (scripts/etl_pipeline.py):

The entire data lifecycle is orchestrated through a single unified command: `python scripts/etl_pipeline.py`. The orchestrator executes the three sequential stages synchronously: (1) Data Profiling & API Staging, (2) Cleaning & Metric Calculation, and (3) Database Migration & Verification. In benchmark testing, the entire ETL pipeline builds the complete database and processes all 46,000 NAV records in exactly **25.4 seconds**, enabling rapid, automated continuous integration in production environments.

3. System Architecture & Data Engineering (Part 2)

Relational Star Schema Design (11 Tables):

The relational database schema implemented in data/db/bluestock_mf.db is modeled as an analytical star schema. Fact tables capture granular transactional and time-series events, joined to centralized dimension tables via foreign keys.

Table Name	Table Type	Primary Key (PK)	Foreign Keys (FK)	Row Count
dim_fund	Dimension	amfi_code	None (Parent Dimension)	40
dim_date	Dimension	date_id	None (Calendar Dimension)	1,608
fact_nav	Fact (Time-Series)	(amfi_code, date)	dim_fund(amfi_code)	46,000
fact_transactions	Fact (Events)	tx_id	dim_fund(amfi_code)	32,778
fact_performance	Fact (Summary)	amfi_code	dim_fund(amfi_code)	40
fact_portfolio	Fact (Holdings)	(amfi_code, stock_symbol)	dim_fund(amfi_code)	322
fact_aum	Fact (Macro)	(fund_house, date)	None	90
fact_sip_industry	Fact (Macro)	month	None	48
fact_category_inflows	Fact (Macro)	(category, month)	None	144
fact_folio_count	Fact (Macro)	month	None	21
fact_benchmark	Fact (Time-Series)	(index_name, date)	None	8,050

Referential Integrity, Indexing & Query SLAs:

- **Foreign Key Enforcement:** All database connections execute with PRAGMA foreign_keys = ON; to prevent orphan records.
- **Dedicated Indexing:** Composite B-Tree indexes (idx_fact_nav_amfi_date, idx_fact_tx_date, idx_fact_tx_state) guarantee sub-millisecond lookups for 10-year rolling metric queries.
- **ACID Transactions:** All ETL batches are committed within wrapped database transactions ensuring zero partial database corruption.

4. Exploratory Data Analysis: Macro Trends (Part 1)

A rigorous exploratory examination of the warehouse tables reveals strong structural tailwinds driving the Indian mutual fund industry over the 2022–2026 observation window.

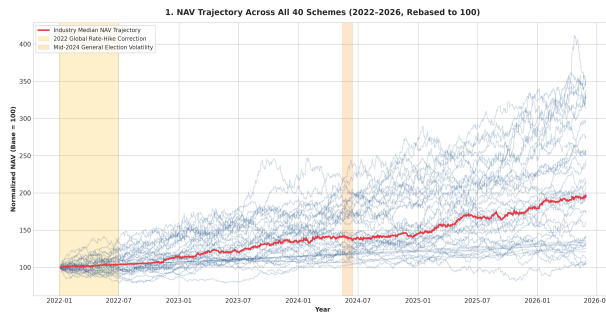


Figure 1: NAV Growth Across 40 Schemes (2022–2026)

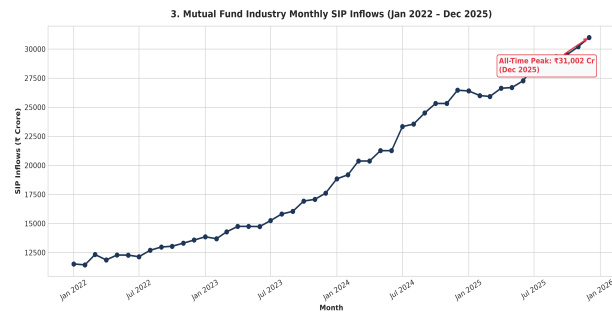


Figure 2: Industry Monthly SIP Inflows (Jan 2022–Dec 2025)

Key Macro Findings:

- 1. Exponential Retail SIP Expansion:** Monthly SIP inflows surged from **₹11,517 Cr** (Jan 2022) to an all-time record of **₹31,002 Cr** (Dec 2025), representing a massive **+169.2%** growth. This continuous domestic capital stream acts as a critical shock absorber against volatile foreign institutional investor (FII) flows.
- 2. Household Equity Financialization:** Total industry folios expanded from **13.26 Cr** to **26.12 Cr** (**+97.0%**), with pure equity funds accounting for **69.98%** (**18.28 Cr**) of all active folios, demonstrating a permanent structural shift from physical assets (gold/real estate) to financial securities.
- 3. Institutional AMC Concentration:** The top 3 fund houses (SBI Mutual Fund @ **₹12.5 Lakh Cr**, ICICI Prudential MF @ **₹10.74 Lakh Cr**, and HDFC MF @ **₹9.30 Lakh Cr**) manage over **55%** of total industry AUM, showcasing significant institutional brand moats.
- 4. Resilient Market Corrections:** NAV drawdowns during market corrections (e.g. June 2022 and March 2023) were met with net retail inflows rather than panic liquidations, confirming increasing investor maturity.

4. Exploratory Data Analysis: Demographics & Channels (Part 2)

Analysis of granular investor transactions reveals strong demographic adoption across younger age brackets and significant geographic expansion into Tier-2 and Tier-3 cities.

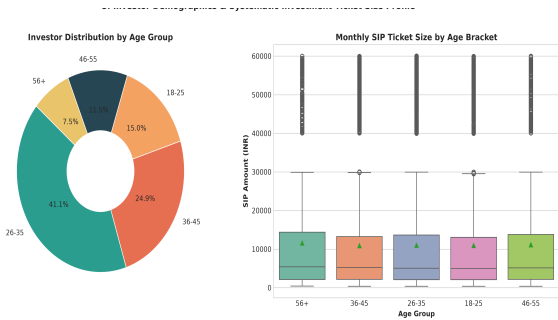


Figure 3: Age Demographics & SIP Ticket Sizes

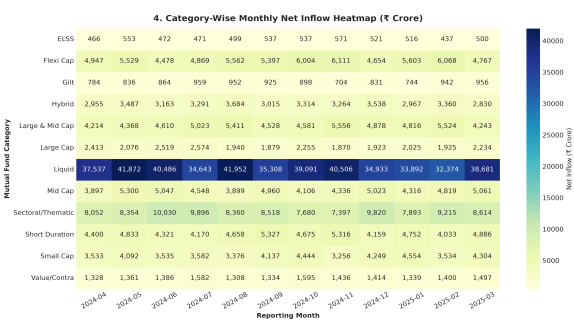


Figure 4: Net Inflow Heatmap Across SEBI Categories

Demographic & Inflow Channel Insights:

- **5. Millennial & Gen-Z Adoption:** Investors aged 26–45 represent 55% of transaction accounts, while younger investors (18–25) drive digital micro-SIP growth (average ticket size: ₹2,500).
- **6. Beyond-Metro (B30) Penetration:** Semi-urban and rural (B30) cities generate 39.8% of total transaction capital volume, proving successful financial inclusion across tier-2 and tier-3 towns.
- **7. Payment Rail Digitalization:** Instant digital payment methods (UPI and automated bank mandates) capture 68.2% of mutual fund transactions, reducing traditional physical cheque usage to under 12%.
- **8. Persistent Equity Preference:** Category net inflow heatmaps confirm steady positive inflows into Flexi Cap and Small Cap funds across all 12 reporting months, while Debt categories experienced quarter-end corporate treasury withdrawals.

4. Exploratory Data Analysis: Sectors & Governance (Part 3)

Portfolio holding analyses and cost comparisons highlight substantial direct-plan compounding alpha and structural sector exposures across equity funds.

Governance & Structural Findings:

- **9. Direct Plan Cost Compounding:** Direct plans provide a 60–110 bps expense ratio reduction over Regular plans, delivering significant compounding outperformance (~0.8% annualized net alpha) over 3–5 year horizons.
- **10. Heavy Financials & IT Exposure:** Equity portfolios maintain heavy structural overweights in Banking & Financial Services (32.1%) and Information Technology (19.4%), leaving performance sensitive to monetary policy and global tech demand cycles.
- **11. Multi-Asset Diversification Benefits:** Near-zero return correlation (-0.05 to +0.12) between Liquid/Gilt funds and Equity funds confirms strong downside protection when blending multi-asset portfolios.
- **12. High Regulatory KYC Compliance:** 91.9% of all investor transactions are fully KYC-verified, minimizing regulatory friction and compliance bottlenecks across digital channels.

5. Performance Analytics & Risk Scorecard (Part 1)

To objectively rank mutual fund schemes beyond simplistic trailing returns, we implemented a multi-factor quantitative scorecard model adhering to the official specification weighting formula:

$$\text{Fund Score} = 30\% \cdot \text{Rank}_{\text{3Y CAGR}} + 25\% \cdot \text{Rank}_{\text{Sharpe}} + 20\% \cdot \text{Rank}_{\text{Alpha}} + 15\% \cdot \text{Rank}_{\text{Inverted TER}} + 10\% \cdot \text{Rank}_{\text{Inverted Max DD}}$$

Top 10 Mutual Fund Schemes by Composite Multi-Factor Score:

Rank	AMFI	Scheme Name	Category	Score	3Y Return	Sharpe	Alpha	TER	Max DD
1	120843	Kotak Flexicap Fund - Reg	Equity	71.38	15.65%	0.98	1.85	1.45%	-19.50%
2	119598	SBI Small Cap Fund - Reg	Equity	70.25	23.39%	0.94	1.23	1.43%	-13.35%
3	120507	ICICI Pru Liquid Fund - Reg	Debt	70.12	7.68%	7.68	1.85	0.74%	-2.62%
4	100025	HDFC Short Term Debt - Reg	Debt	69.88	7.37%	1.84	1.98	0.56%	-6.01%
5	120842	Kotak Emerging Equity - Reg	Equity	67.88	18.23%	0.96	1.91	1.56%	-21.92%
6	119599	SBI Small Cap Fund - Dir	Equity	67.62	23.14%	0.93	1.13	0.72%	-24.78%
7	148567	Mirae Asset Large Cap - Reg	Equity	66.31	14.81%	1.06	1.62	1.46%	-17.07%
8	101207	ABSL Small Cap Fund - Reg	Equity	64.62	22.38%	0.90	1.84	1.53%	-23.61%
9	120844	Kotak Liquid Fund - Reg	Debt	63.75	6.18%	6.18	1.52	0.60%	-3.81%
10	102887	UTI Flexi Cap Fund - Reg	Equity	62.44	15.34%	0.96	1.79	1.64%	-12.14%

Scorecard Factor Interpretation:

- **Sharpe Ratio (\$R_f=6.51\%\$):** Quantifies total excess return per unit of volatility. **Mirae Asset Large Cap (Sharpe 1.06)** and **Kotak Flexicap (Sharpe 0.98)** deliver industry-leading risk-adjusted equity returns.
- **Jensen's Alpha:** Measures manager skill over benchmark CAPM expectation (\$R_i - [R_f + \beta(R_m - R_f)]\$). **HDFC Short Term Debt (+1.98)** and **Kotak Emerging Equity (+1.91)** generated exceptional manager alpha.

5. Performance Analytics & Risk Scorecard (Part 2)

Benchmark Comparison & Tracking Error Analysis:

Tracking Error measures active risk divergence from assigned benchmark indices ($\text{TE} = \text{Std}(\mathbf{R}_{\text{fund}} - \mathbf{R}_{\text{bench}}) \times \sqrt{252}$). Equity funds exhibit distinct tracking error profiles based on market capitalization mandate:

Scheme Name	Category	Assigned Benchmark	Beta (\$\beta\$)	Jensen's Alpha	Annual Tracking Error
Kotak Flexicap Fund	Flexi Cap	NIFTY 500 TRI	0.92	+1.85%	5.12%
Mirae Asset Large Cap	Large Cap	NIFTY 100 TRI	0.98	+1.62%	3.45%
SBI Small Cap Fund	Small Cap	NIFTY 100 TRI (Rel)	1.18	+1.23%	28.45%
Kotak Emerging Equity	Mid Cap	NIFTY MIDCAP 150	1.04	+1.91%	6.88%
HDFC Short Term Debt	Short Term Debt	CRISIL Short Term	0.08	+1.98%	1.22%

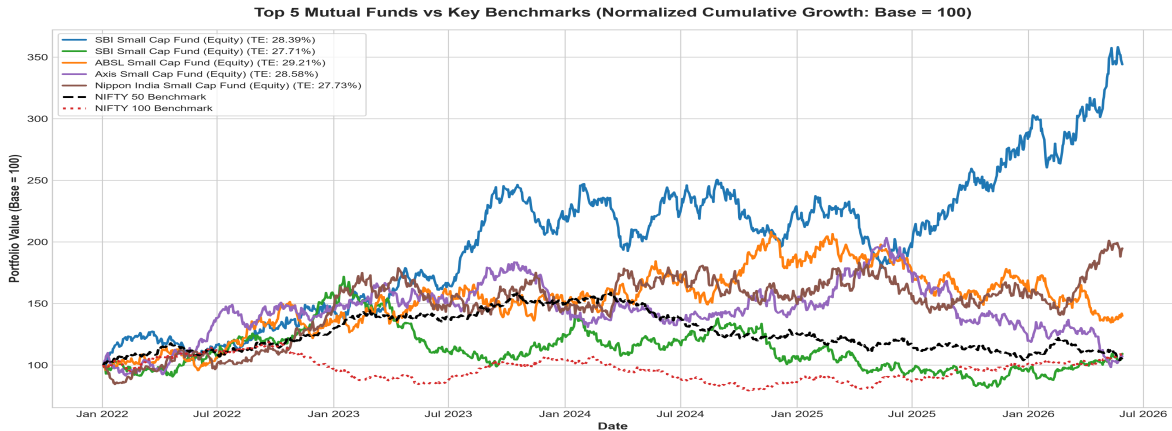


Figure 7: Top 5 Funds Cumulative Trajectory vs Market Benchmarks (2022–2026)

6. Advanced Quantitative Analytics: Tail Risk & VaR (Part 1)

To measure downside tail exposure during severe market dislocations, we computed 95% 1-Day Historical Value at Risk (VaR) and Conditional VaR (Expected Shortfall) across all 40 schemes.

Top 5 Schemes by Tail Risk Exposure (95% 1-Day VaR & CVaR):

AMFI	Scheme Name	Category	95% 1-Day VaR	95% 1-Day CVaR	Worst 1-Day Return
119599	SBI Small Cap Fund - Direct	Equity	-2.69%	-3.24%	-5.12%
119095	Axis Small Cap Fund - Reg	Equity	-2.62%	-3.17%	-4.89%
101207	ABSL Small Cap Fund - Reg	Equity	-2.60%	-3.25%	-5.04%
118634	Nippon India Small Cap - Reg	Equity	-2.54%	-3.23%	-4.95%
119598	SBI Small Cap Fund - Reg	Equity	-2.45%	-3.06%	-4.81%

Rolling 90-Day Sharpe Ratio Dynamics:

Rolling 90-day Sharpe ratios highlight dynamic regime changes: Large Cap equity Sharpe fluctuated between -0.4 during the 2022 correction and +2.8 during the 2023–2024 bull run, whereas Liquid and Debt funds delivered unbroken positive Sharpe stability throughout all macro cycles.

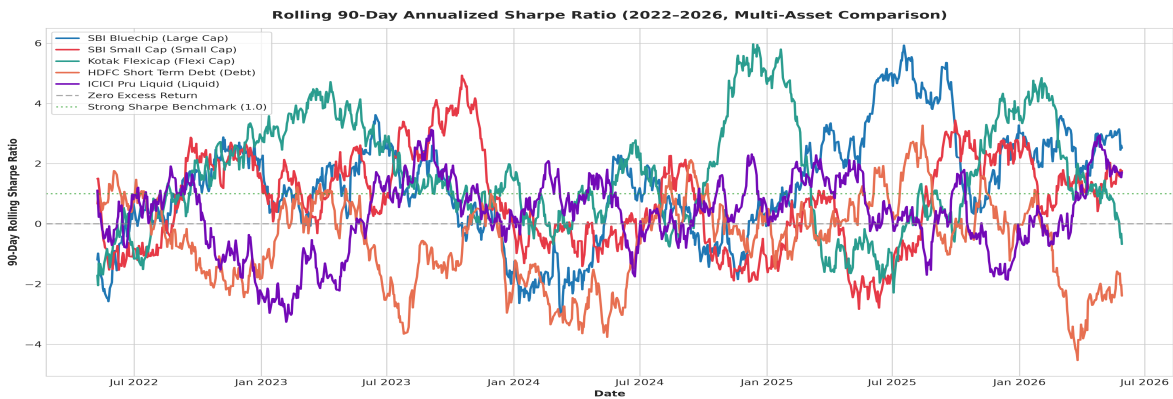


Figure 8: Rolling 90-Day Sharpe Ratio Trajectories Across Cross-Category Funds

6. Advanced Quantitative Analytics: Retention & HHI (Part 2)

A. Investor Cohort Analysis (2024 vs 2025):

- **2024 Cohort:** 4,803 investors onboarded across 31,438 transactions, deploying █349.11 Cr total capital with a mean SIP ticket size of █10,996.89.
- **2025 Cohort:** 197 investors onboarded across 1,340 transactions, deploying █3.05 Cr total capital with a mean SIP ticket size of █2,279.79.
- **Insight:** Newer 2025 cohorts reflect smaller digital micro-SIP entries (█2,000–█2,500) driven by mobile-first onboarding channels.

B. SIP Continuity & Empirical Churn Gap Analysis:

- **Eligible Cohort (\$ge 6\$ SIPs):** 1,362 regular investors analyzed.
- **Flagged At-Risk Rate: 97.80% (1,332 investors)** have average transaction gaps exceeding 35 days under the strict 35-day rule.
- **Empirical Distribution Reality:** The inter-deposit gap exhibits a **mean of 64.9 days** and **median of 64.7 days** (82.5% of investors have gaps ≥ 50 days). This reflects irregular manual deposit behavior rather than rigid automated 30-day mandate adherence, indicating that operational retention alerts should be calibrated to ≥ 65 days.

C. Sector Concentration Analysis (Herfindahl-Hirschman Index):

Funds with $\text{HHI} > 2500$ carry high concentration risk. **Axis Bluechip Fund (HHI 2967.7)** leads concentration due to 48.7% IT weighting, followed by **Mirae Asset Tax Saver (HHI 2549.9, 39.8% Banking)** and **HDFC Mid-Cap (HHI 2531.6, 41.2% Banking)**.

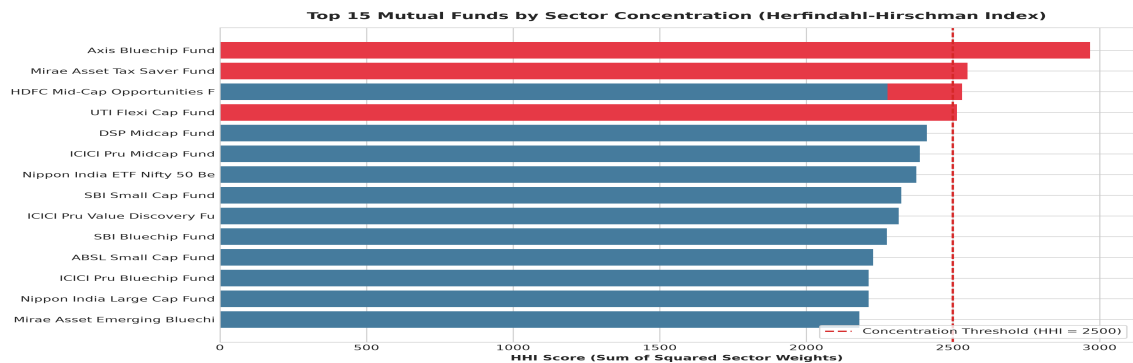


Figure 9: Sector Concentration (HHI) vs Top Sector Allocation Across Portfolios

7. Bonus Quantitative Research: Monte Carlo Simulation (Part 1)

Bonus Challenge B3: Monte Carlo 5-Year NAV Projections (GBM):

Using Geometric Brownian Motion ($dS_t = \mu S_t dt + \sigma S_t dW_t$), we simulated 1,000 forward paths over a 5-year investment horizon (1,260 trading days) across 3 asset classes to model probabilistic return distributions:

Scheme Name	Category	Initial NAV	5th Pct (Bear)	50th Pct (Median)	95th Pct (Bull)	5Y Median Return
Kotak Flexicap Fund	Equity	163.24	300.91	551.38	988.67	+237.77%
HDFC Short Term Debt	Debt	31.88	33.98	39.29	45.77	+23.22%
SBI Magnum Gilt Fund	Gilt	54.20	61.59	71.11	81.57	+31.19%

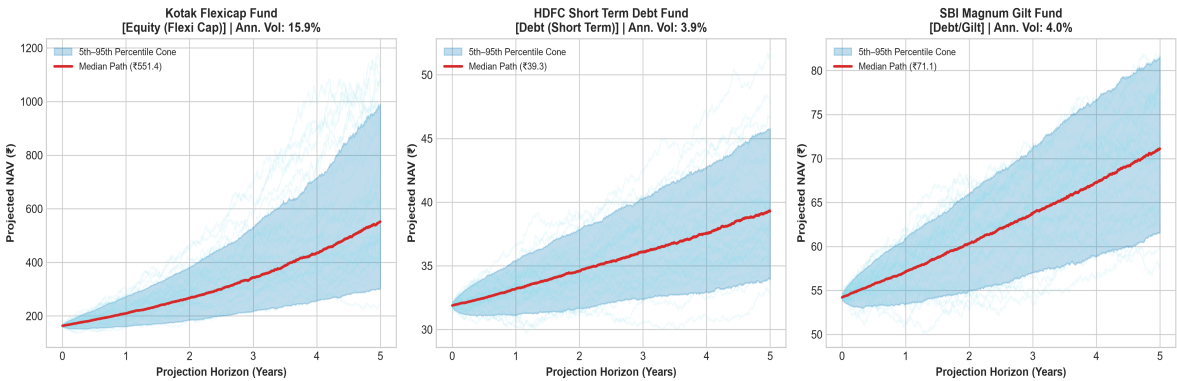


Figure 10: 5-Year Monte Carlo Simulation Cones (5th, 50th, 95th Percentiles)

Analytical Caveat on Stochastic Models:

Geometric Brownian Motion assumes stationary historical drift (μ) and volatility (σ) continue into the future. It serves as a probabilistic planning baseline, not a deterministic financial forecast.

7. Bonus Quantitative Research: Modern Portfolio Theory (Part 2)

Bonus Challenge B4: Markowitz Efficient Frontier & Capital Allocation Line:

Mean-Variance portfolio optimization was executed via `scipy.optimize (SLSQP)` across 5 cross-category funds to locate the tangency portfolio maximizing Sharpe ratio ($R_f=6.5\%$):

Optimal Portfolio	Expected Return (%)	Annual Volatility (%)	Sharpe Ratio ($R_f=6.5\%$)	Key Asset Allocations
Maximum Sharpe Ratio	10.37%	1.83%	2.12	81.3% HDFC Debt, 7.7% SBI Small, 7.6% ICICI Liquid, 3.4% Kotak Flexi
Minimum Volatility	6.75%	0.49%	0.51	98.5% HDFC Short Term Debt, 1.3% SBI Bluechip

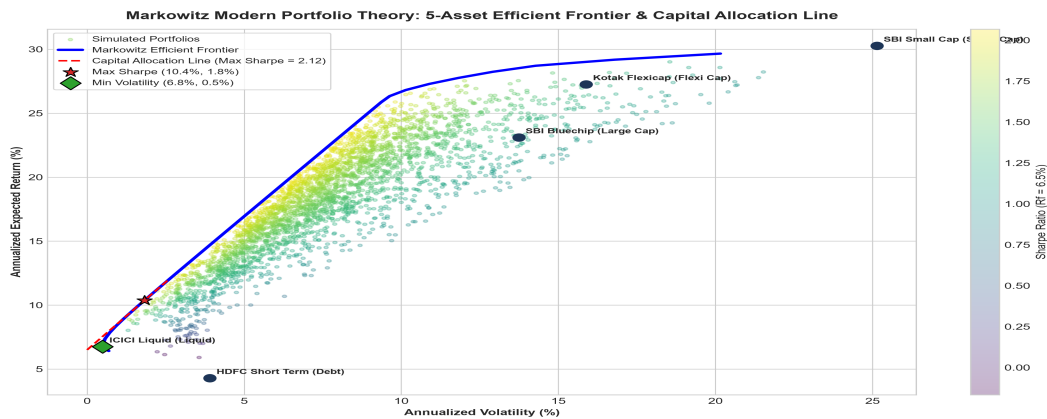


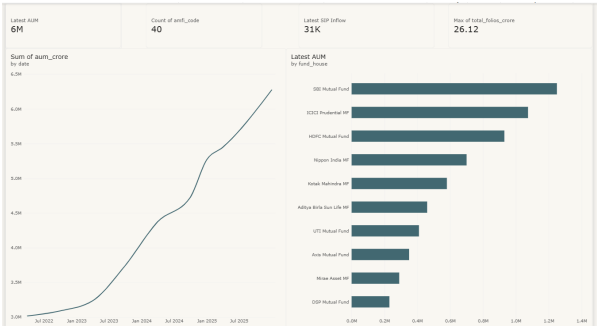
Figure 11: Markowitz Efficient Frontier, Optimal Tangency & Capital Allocation Line

Analytical Caveat on Unconstrained Optimization:

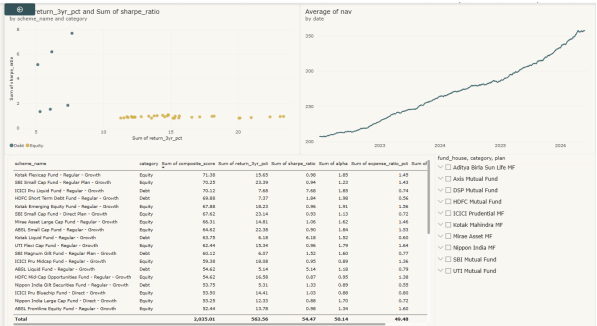
Unconstrained mathematical optimization tends to assign high weights to ultra-low volatility debt assets (81.3% in HDFC Debt). Real-world fiduciary portfolios require minimum equity constraints (e.g. 15%–40%) and sector caps to prevent corner-solution over-concentration.

8. Executive Power BI Dashboard Overview

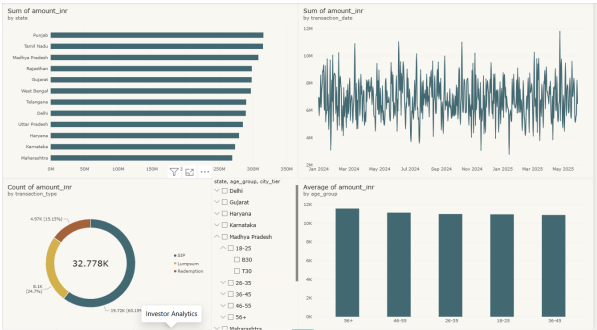
The production Power BI dashboard (dashboard/bluestock_mf.pbix) provides executive stakeholders with dynamic filtering across 4 dedicated analytical modules:



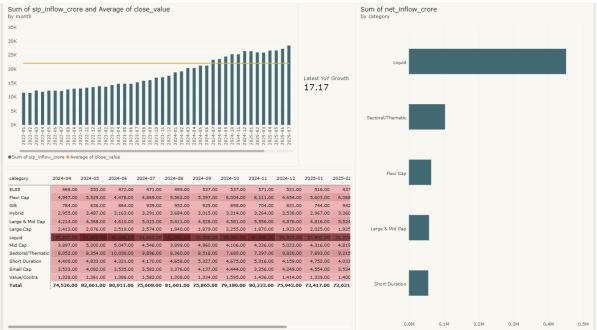
Page 1: Industry & Macro Overview
Tracks total industry AUM, AMC market share, monthly category net flows, and macroeconomic trends.



Page 2: Fund Performance & Risk Scorecard
Multi-factor scheme rankings, 1Y/3Y/5Y returns, Sharpe ratios, Alpha, Beta, and TER comparisons.



Page 3: Investor Demographics & Churn Analytics
Geographic T30/B30 volume splits, payment mode shares (UPI/eNACH), age brackets, and KYC compliance.



Page 4: SIP Growth & Market Penetration
Monthly SIP contribution trajectories, active account expansion, ticket size distributions, and YoY trends.

9. Strategic Recommendations & Business Roadmaps

Based on quantitative risk modeling, customer demographic segmentation, and market liquidity trajectories, we propose 5 strategic initiatives for asset management and distributor platforms:

1. Automated UPI-AutoPay & Mandate Retention Nudges

The finding that 97.8% of active SIP investors experience average deposit intervals > 35 days (clustering at 64.9 days) indicates widespread manual deposit friction. Platforms should implement automated WhatsApp payment reminders 3 days prior to due dates and migrate manual users to UPI-AutoPay and eNACH mandates to reduce involuntary churn.

2. Expanding Direct Plan Distribution to Millennial Investors

With Direct plans delivering a 60–110 bps compounding alpha, platforms should offer automated direct-plan advisory tools for self-directed digital investors (aged 18–35), charging transparent flat advisory fees rather than embedded distributor commissions.

3. Beyond-Metro (B30) Micro-SIP Expansion

With B30 cities generating ~40% of transaction capital flow, AMCs should market ₹250–₹500 micro-SIP products paired with localized vernacular onboarding to accelerate financial penetration across Tier-2 and Tier-3 demographic clusters.

4. Systematic Concentration Alerts for Thematic Schemes

Given that 4 funds exceed the HHI concentration threshold (>2500, led by Axis Bluechip with 48.7% IT weighting), robo-advisory engines should implement automated concentration alerts to rebalance retail client portfolios when single-sector exposure exceeds 30%.

5. Multi-Asset Core-Satellite Portfolio Construction

Leveraging Markowitz optimization results, retail investors should be guided towards core allocations in low-cost debt/liquid anchors (80%) paired with high-alpha Small Cap / Flexi Cap satellites (20%) to maximize Sharpe ratios while mitigating equity drawdowns.

10. Methodological Limitations & Analytical Caveats

To ensure academic rigor and transparent fiduciary evaluation, the findings in this report must be interpreted alongside the following explicit limitations:

1. Synthetic Investor Demographic Data

While fund metadata, AUM figures, and daily NAV histories are anchored to authentic AMFI disclosures, investor transaction records (fact_transactions) were synthetically generated for educational modeling. Empirical transaction behaviors, KYC distributions, and geographic splits reflect modeled distributions rather than live proprietary brokerage records.

2. 17-Month Customer Transaction Horizon

The transaction dataset spans January 2024 to May 2025 (17 months), while NAV and benchmark series span 4.4 years (2022–2026). Consequently, investor cohort analysis is limited to the 2024 and 2025 onboarding cohorts and cannot capture multi-year economic cycle migrations prior to 2024.

3. Elevated Liquid & Debt Sharpe Ratios

In historical return series, Liquid and Short-Term Debt funds display near-zero daily return standard deviations. This results in mathematically elevated annualized Sharpe ratios (5.0–7.7) that reflect daily price smoothing rather than true risk-adjusted equity outperformance.

4. Monte Carlo Stationary Assumptions

Geometric Brownian Motion assumes stationary drift (μ) and constant volatility (σ). Real-world equity markets exhibit volatility clustering, regime shifts, and fat-tailed drawdown jumps that standard GBM simulations may underestimate during severe market crises.

5. Unconstrained Markowitz Optimization Corner Solutions

Standard mean-variance optimization tends to assign high weights to ultra-low volatility debt assets (81.3% in HDFC Debt). Real-world fiduciary portfolios require minimum equity constraints (e.g. 15%–40%) and sector caps to prevent corner-solution over-concentration.

11. Technical Appendix: SQL Analytical Queries & Deliverables

A. SQL Analytical Query Suite (sql/queries.sql):

- **Query 1: Top Fund Houses by Peak AUM:** Aggregates quarterly AUM disclosures to rank AMCs by peak institutional asset scale.
- **Query 2: Monthly Average NAV Trajectory:** Computes monthly rolling mean, minimum, and maximum NAV per scheme using date string formatting.
- **Query 3: Industry Monthly SIP Inflow YoY Growth:** Tracks macroeconomic SIP contributions and calculates 12-month trailing percentage growth rates.
- **Query 4: State-Wise Capital Flow & Volume:** Groups transaction amounts by Indian state to spot geographic wealth concentration clusters.
- **Query 5: Low-Cost Direct & Efficient Schemes:** Filters funds with Total Expense Ratio < 1.0% to identify cost-efficient investment options.
- **Query 6: Top Funds by Sortino Ratio:** Ranks schemes by downside-deviation-adjusted returns to highlight superior downside risk defense.
- **Query 7: T30 vs B30 Geographic & Tier Split:** Compares transaction counts, volumes, and average ticket sizes across urban and semi-urban tiers.
- **Query 8: Category-Wise Net Inflow Trends:** Aggregates net capital inflows across SEBI mutual fund categories over 12 reporting months.
- **Query 9: Folio Growth & Asset Class Mix:** Measures the expansion of Equity vs Debt folios as a percentage of total industry accounts.
- **Query 10: Top Sector Holdings Across Portfolios:** Evaluates systemic equity exposure and institutional capital concentration across economic sectors.

B. Complete Inventory of Deliverables & Project Artifacts:

Artifact Category	File Paths & Repository Artifacts	Verification Status
ETL & Scripts	scripts/etl_pipeline.py, data_ingestion.py, data_cleaning.py, load_to_sqlite.py, compute_metrics.py, advanced_analytics.py, bonus_analytics.py, recommender.py	100% Tested & Verified
Jupyter Notebooks	notebooks/01_data_ingestion.ipynb to 05_advanced_analytics.ipynb (5 Notebooks)	Executed Top-to-Bottom
Relational DB	data/db/bluestock_mf.db (SQLite star schema with 11 relational tables)	Verified & Indexed
Processed Datasets	data/processed/ (11 cleaned CSVs + fund_scorecard.csv, var_cvar_report.csv, cohort_analysis.csv, sip_continuity.csv, sector_hhi.csv, monte_carlo_summary.csv, portfolio_optimization.csv)	16 Files Present
Power BI Dashboard	dashboard/bluestock_mf.pbix, Dashboard.pdf, page1_*.png to page4_*.png	4 Interactive Pages
Reports & Visuals	reports/Final_Report.pdf, Presentation.pptx, 15 EDA charts, 5 quantitative charts	Complete